Cross-lingual Metrics: Evaluating Machine Translation without Human Reference

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Summary

+ Contextualized embeddings push semantic similarity metrics to a new level
- Reference-free metrics remain a challenge

Our contributions:

- Discuss similarity metrics in the cross-lingual context
- Contemplate the formulation of text quality metric
- A combination of metrics and fluency measures yields a good overall metric without any reference required

Lexical Similarity (Bilingual Dictionary)

- Full credit for matching cross-lingual aligned words:

  SYSTEM Jim went home

  SOURCE Jim ging nach Hause
- Partial credit for matching cross-lingual stems:

  SYSTEM Jim went home

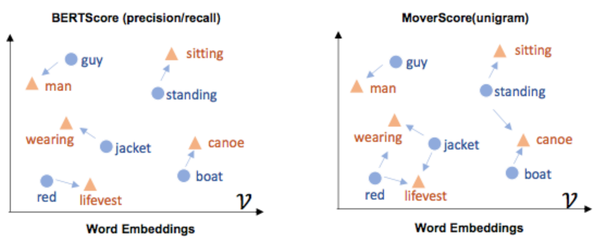
  SOURCE Jim geht nach Hause
- Partial credit for matching cross-lingual synonyms:

  SYSTEM Jim walks home

  SOURCE Jim geht nach Hause

Semantic Similarity (Soft and Hard Alignment)

- Compute the semantic similarity of a sentence pair based on their cross-lingual embeddings with the alignments below:



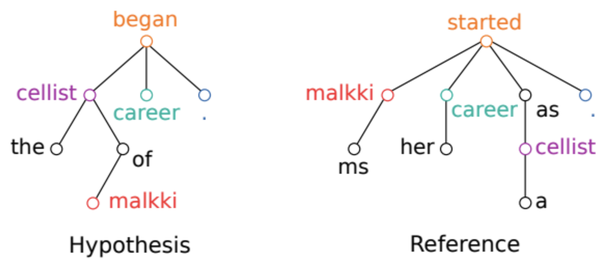
System x: A **guy** with a **red jacket** is **standing** on a **boat**

Ref y: A **man wearing** a **lifevest** is **sitting** in a **canoe**

Figure 1: An illustration of MoverScore and BERTScore.

Syntactic Similarity (Dependency Tree)

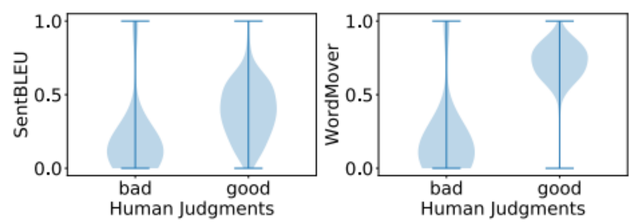
- Parse a sentence pair into the dependency trees below:



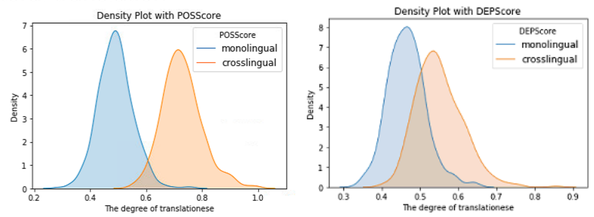
- Tree Edit Distance (Zhang and Shasha 1989) : count the number of delete, insert and match/substitution operations needed to turn one tree into another

Discussion on Similarity

- Lexical Similarity: light-weight, fast to compute, interpretable and focus on lexis
- Semantic similarity: a) strong in distinguishing high-quality outputs from mediocre and low-quality outputs b) achieve substantially better performance than lexical similarity metrics



- Syntactic Similarity: effective in diagnosing the degree of literal translation



Fluency Measure

How likely is a string of translation words good translation?

Helps for order:

$$p_{\text{LM}}(\text{the house is small}) > p_{\text{LM}}(\text{small the is house})$$

Helps for lexical choice:

$$p_{\text{LM}}(\text{I am going home}) > p_{\text{LM}}(\text{I am going house})$$

Cross-lingual Text Quality Metric

How to choose a metric in the reference-free context?

Text Quality Metric = Lexical Similarity? Semantic Similarity? Syntactic Similarity? Fluency? A combination thereof?

Settings	cs-en	de-en	fi-en	lv-en	ru-en	tr-en	zh-en
Monolingual Metrics							
SentBLEU	0.435	0.432	0.571	0.393	0.484	0.538	0.512
chrF++	0.523	0.534	0.678	0.520	0.588	0.614	0.593
METEOR++	0.552	0.538	0.720	0.563	0.627	0.626	0.646
MoverScore	0.670	0.708	0.835	0.746	0.738	0.762	0.744
Cross-lingual Metrics							
MoverScore + XLM	0.069	0.331	0.045	0.006	0.213	0.034	0.227
MoverScore + MBERT	0.282	0.370	0.310	0.182	0.228	0.310	0.452
MoverScore + LASER	0.325	0.401	0.413	0.482	0.363	0.422	0.466
XScore	0.374	0.494	0.464	0.474	0.393	0.483	0.542
Language Model							
OpenAI-GPT	0.312	0.255	0.522	0.454	0.357	0.276	0.269
OpenAI-GPT2	0.352	0.343	0.569	0.496	0.394	0.336	0.343
Language Model + Cross-lingual Metrics							
XMoverScore	0.468	0.486	0.612	0.642	0.461	0.543	0.560

Table 1: Pearson correlations with segment-level human judgments.

Closing thoughts

Takeaways:

- Text Quality != Semantic Similarity
- Desirable Metric is supposed to take into account lexicality, semantics, syntactics and fluency
- A combination thereof yields a nice overall metric in the cross-lingual context

References

1. MoverScore: Text Generation Evaluating with Contextualized Embeddings and Earth Mover Distance, EMNLP, 2019
2. Simple Fast Algorithms for the Editing Distance Between Trees and Related Problems, SIAM Journal on Computing, 1989